



Paper Review

Structural Causal Bandit : Where to Intervene?

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Preliminary : Multi-Armed Bandit

Introduction to MAB, and Regret

Multi-Armed Bandit (MAB)

Problem Definition

- ▶ Maximizing cumulated reward under uncertainty
 - $\mathcal{A} = \{a_1, a_2, \dots, a_K\}$
 - Each Arm's Reward is independent to each other (original setting)
- ▶ Regret
 - Cumulative reward difference between optimal arm and selected arm
 - $$R_T = T\mu^* - \sum_{t=1}^T \mu_{a_t}$$

T : total number of trials

μ^* : expected reward of the optimal arm

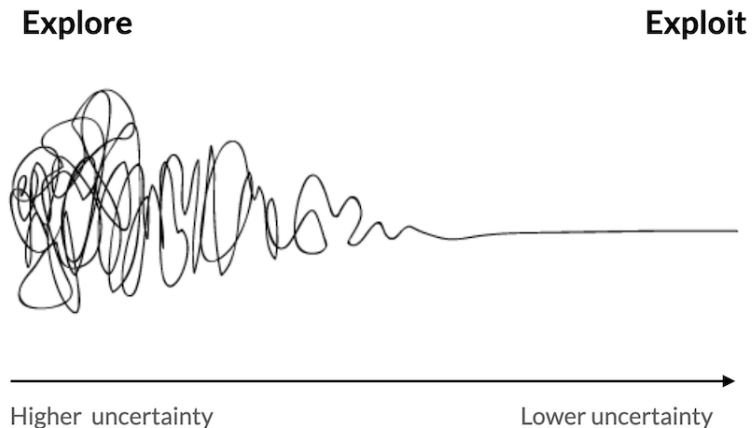
μ_{a_t} : expected reward of the arm chosen at time t



Multi-Armed Bandit (MAB)

Exploration & Exploitation Trade-off

- ▶ We want to know the best arm
 - Many different strategies exist to find the best arm
 - Merely finding the best is not enough
- ▶ Trade-off
 - Exploration : Removing Uncertainty (Acquisition of Information)
 - Exploitation : Choosing the best arm we know (Greedy Selection)



Multi-Armed Bandit (MAB)

Regret Growth Rate

► ϵ -greedy

- We select the “best” arm with probability $1 - \epsilon$, and a random arm with probability ϵ
- Fixed ϵ : $\mathbb{E}[R_T] \approx \epsilon \cdot \delta \cdot T$

δ : Expected reward gap between the optimal and the suboptimal arm

Algorithm	Expected Regret	Main Idea
Random	$O(T)$	Pure exploration
ϵ -greedy	$O(\epsilon T), O(\log T)$	Fixed exploration rate
UCB	$O(\log T)$	Optimistic exploration
Thompson Sampling	$O(\log T)$	Bayesian Posterior

Non-Trivial Dependencies Among Arms

Introduction to Linear Bandit, and SCM-MAB

Non-trivial dependencies among arms

Linear Bandit

► Reward Function

- $Reward_i = \theta^T x_i + \epsilon$ (Shape Convention)
- Now reward of arm gives you information for other arms

► Learning $\hat{\theta}$ = Learning Expected Value for Arms Simultaneously

- LinUCB : UCB-variant for Linear Bandit
- Regret for LinUCB, and UCB
- LinUCB is better if there are lots of arms : $d < \sqrt{K}$

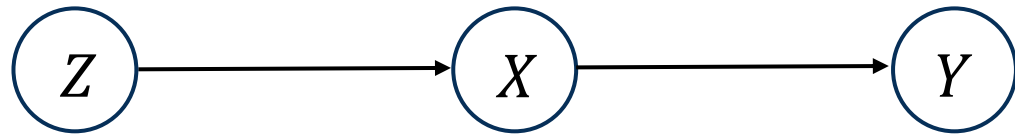
Algorithm	Gap-dependent Regret	Gap-free Regret
UCB	$O\left(\sum_{a:\Delta_a>0} \frac{\log T}{\Delta_a}\right)$	$O(\sqrt{KT \log T})$
LinUCB	$O\left(\frac{d \log T}{\Delta_{\min}}\right)$	$O(d\sqrt{T \log T})$

Non-trivial dependencies among arms

SCM-MAB : MAB with Structural Causal Model

► Action = Intervention

- Pulling Arm in SCM-MAB is same with do-operation $\text{do}(\mathbf{X}=\mathbf{x})$
- Due to graph surgery, $P(y|\text{do}(X), \text{do}(Z)) = P(y|\text{do}(X))$
- Skip trials for $\text{do}(X=0, Z=0)$, $\text{do}(X=0, Z=1)$ but $\text{do}(X=0)$



Possible Actions

$\text{do}(\emptyset)$

$\text{do}(X = 0), \text{do}(X = 1)$

$\text{do}(Z = 0), \text{do}(Z = 1)$

$\text{do}(X = 0, Z = 0), \text{do}(X = 0, Z = 1), \text{do}(X = 1, Z = 0), \text{do}(X = 1, Z = 1)$

SCM-MAB

Skipping Suboptimal Arms using Properties of Causal Diagram

Equivalence Among Arms

Recap : Do-Calculus

Theorem (Rules of Do-calculus Pearl [1995])

The following transformations are valid for any do-distribution induced by a causal model $\mathcal{M}$:

Rule 1: *Adding/removing Observations*

$$P(\mathbf{y} | do(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} | do(\mathbf{x})) \quad \text{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}.$$

Rule 2: *Action/observation Exchange*

$$P(\mathbf{y} | do(\mathbf{x}), do(\mathbf{z})) = P(\mathbf{y} | do(\mathbf{x}), \mathbf{z}) \quad \text{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}.$$

Rule 3: *Adding/removing Actions*

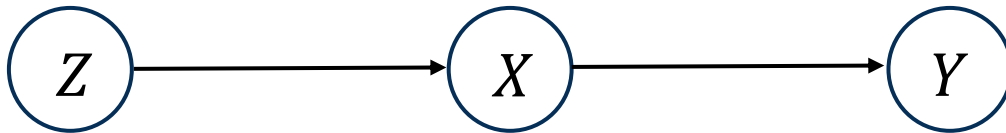
$$P(\mathbf{y} | do(\mathbf{x}), do(\mathbf{z})) = P(\mathbf{y} | do(\mathbf{x})) \quad \text{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}.$$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- Skip Redundant Actions using **Rule 3**

Definition 1 (Minimal Intervention Set (MIS)). A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is said to be a *minimal intervention set* relative to $\llbracket G, Y \rrbracket$ if there is no $\mathbf{X}' \subset \mathbf{X}$ such that $\mu_{\mathbf{x}[\mathbf{X}']} = \mu_{\mathbf{x}}$ for every SCM conforming to the G .



MIS in the Chain

$$do(X, Z) = do(X) \rightarrow \{X\}$$

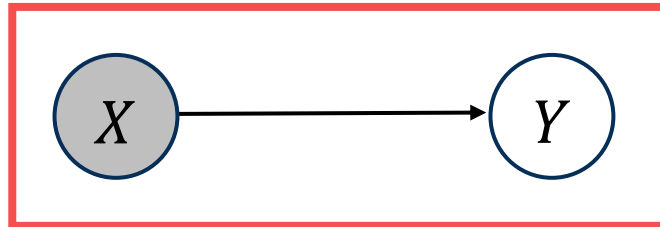
$$\therefore \text{MIS} : \{\emptyset, \{X\}, \{Z\}\}$$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- ▶ Skip Redundant Actions using **Rule 3**

Proposition 1 (Minimality). *A set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a minimal intervention set for G with respect to Y if and only if $\mathbf{X} \subseteq an(Y)_{G_{\overline{\mathbf{X}}}}$.*



MIS in the Chain

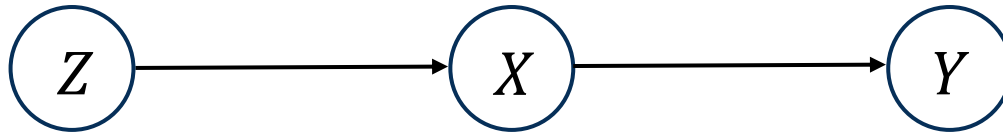
$$do(X, Z) = do(X) \rightarrow \{X\}$$

$$\therefore \text{MIS} : \{\emptyset, \{X\}, \{Z\}\}$$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- ▶ We can skip further with Partial-orders among arms



$$\max_{z \in D(Z)} \mu_z \leq \max_{x \in D(X)} \mu_x$$

$$\mu_z^* = \sum_x \mu_x \cdot P(x|do(z^*)) \leq \sum_x \mu_x^* P(x|do(z^*)) = \mu_x^*$$

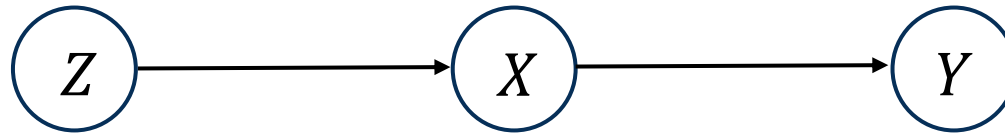
$\therefore \text{do}(X=0), \text{do}(X=1)$: Possibly Optimal Interventions

Equivalence Among Arms

POMIS (Possibly Optimal Minimal Intervention Set)

- Our Goal : Find every POMIS in the given causal diagram

Definition 2 (Possibly-Optimal Minimal Intervention Set (POMIS)). Given information $\llbracket G, Y \rrbracket$, let $\mathbf{X}$ be a MIS. If there exists a SCM conforming to G such that $\mu_{\mathbf{x}^*} > \forall \mathbf{z} \in \mathbb{Z} \setminus \{\mathbf{x}\} \mu_{\mathbf{z}^*}$, where $\mathbb{Z}$ is the set of MISs with respect to G and Y , then $\mathbf{X}$ is a *possibly-optimal minimal intervention set* with respect to the information $\llbracket G, Y \rrbracket$.



We can easily confirm that Z , and empty set is not POMIS, but a X
 $\therefore \text{do}(X=0), \text{do}(X=1)$: Possibly Optimal Interventions

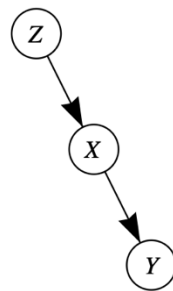
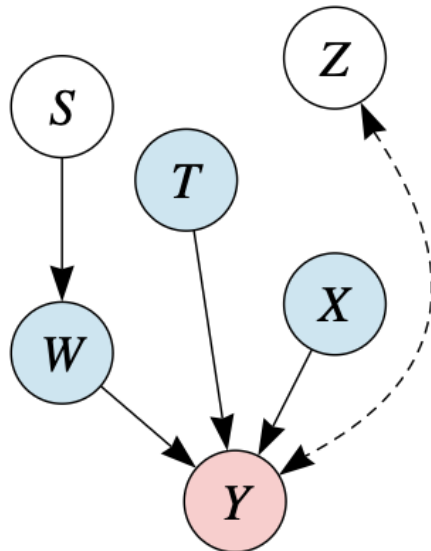
Equivalence Among Arms

POMIS (Possibly Optimal Minimal Intervention Set)

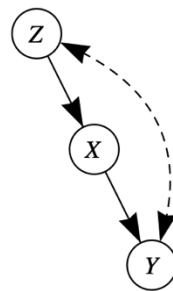
- Our Goal : Find every POMIS in the given causal diagram

Proposition 2. *Given information $\llbracket G, Y \rrbracket$, if Y is not confounded with $an(Y)_G$ via unobserved confounders, then $pa(Y)_G$ is the only POMIS.*

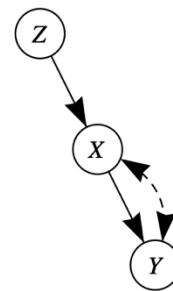
Corollary 3 (Markovian POMIS). *Given $\llbracket G, Y \rrbracket$, if G is Markovian, then $pa(Y)_G$ is the only POMIS.*



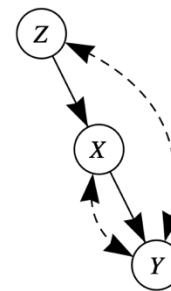
(a)



(b)



(c)



(d)

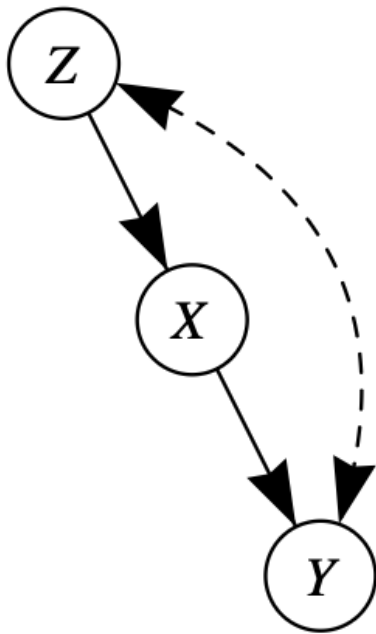
	$\emptyset$	$D(X)$	$D(Z)$
(a)		✓	
(b)	✓	✓	
(c)		✓	✓
(d)	✓	✓	✓

(e)

Equivalence Among Arms

POMISs of Graph with Unobserved Confounder

► Front-Door Case



Assume given variables are binary variables, and $P(U=1) = P(U=0) = 0.5$

$$Z \leftarrow U$$

$$X \leftarrow Z$$

$$Y \leftarrow \neg(X \oplus U)$$

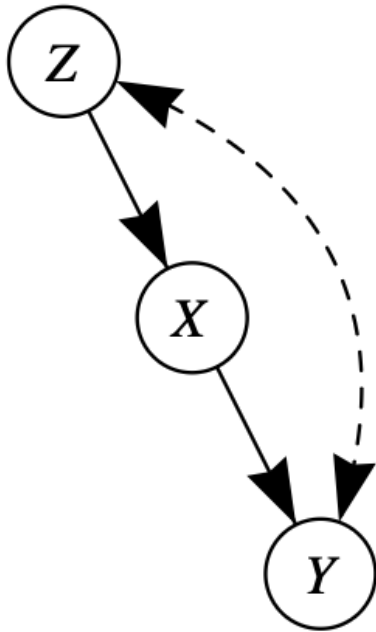
If we intervene Z or X , expected value of Y is 0.5

However, Y is always one if we do nothing

Equivalence Among Arms

POMISs of Graph with Unobserved Confounder

► Front-Door Case



$$\mathbb{E}[y \mid do(z^*)] = \sum_x \mu_x P(x \mid do(z^*)) \leq \sum_x \mu_x^* P(x \mid do(z^*)) = \mu_x^*$$

$\therefore$ POMISs = $\{\emptyset, \{X\}\}$

Q : How can we gain POMISs ?

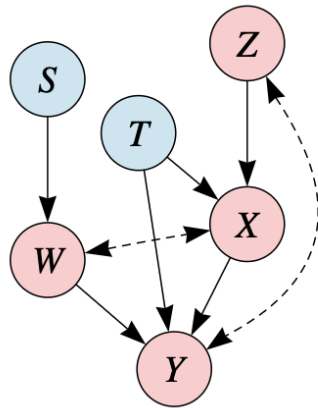
A : Using Graphical Characterization of POMIS

Graphical Characterization of POMIS

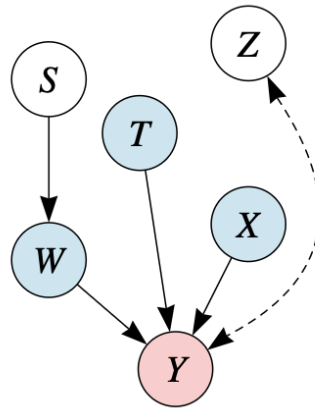
MUCT (Minimal Unobserved-Confounders' Territory)

- Our Goal : Find every POMIS in the given causal diagram

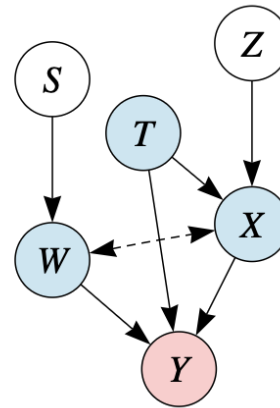
Definition 3 (Unobserved-Confounders' Territory). Given information $\llbracket G, Y \rrbracket$, let H be $G[An(Y)_G]$. A set of variables $\mathbf{T} \subseteq \mathbf{V}(H)$ containing Y is called an *UC-territory* on G with respect to Y if $De(\mathbf{T})_H = \mathbf{T}$ and $CC(\mathbf{T})_H = \mathbf{T}$.



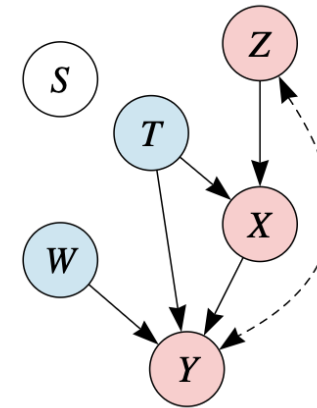
(a) G



(b) $G_{\overline{X}}$



(c) $G_{\overline{Z}}$



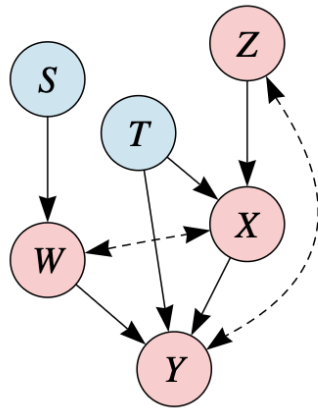
(d) $G_{\overline{W}}$

Graphical Characterization of POMIS

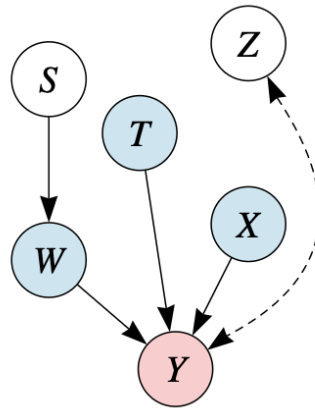
MUCT (Maximal Unobserved-Confounders' Territory)

- Our Goal : Find every POMIS in the given causal diagram

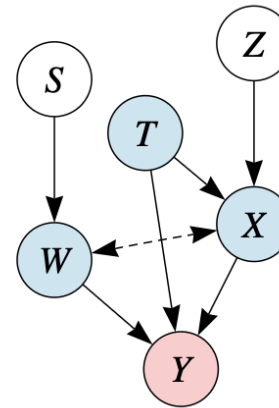
Definition 4 (Interventional Border). Let $\mathbf{T}$ be a minimal UC-territory on G with respect to Y . Then, $\mathbf{X} = pa(\mathbf{T})_G \setminus \mathbf{T}$ is called an *interventional border* for G with respect to Y .



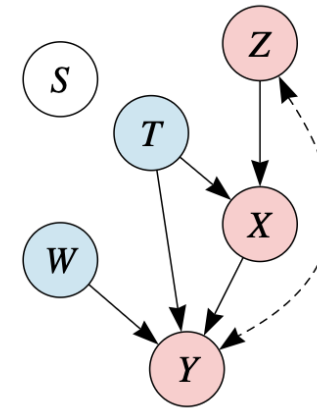
(a) G



(b) $G_{\overline{X}}$



(c) $G_{\overline{Z}}$



(d) $G_{\overline{W}}$

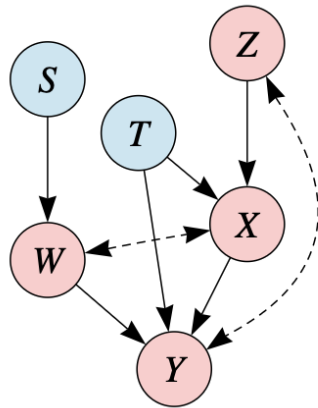
Graphical Characterization of POMIS

Intervention Border : Parents of MUCT

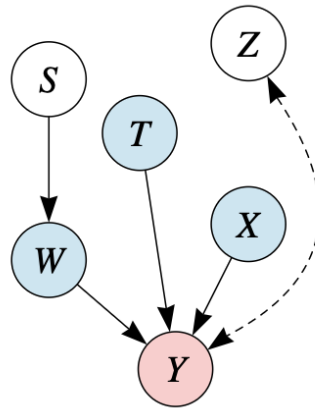
► Finding POMISs = Finding IBs

Proposition 4. $IB(G, Y)$ is a POMIS given $\llbracket G, Y \rrbracket$.

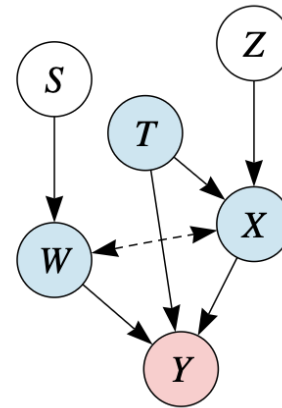
Proposition 5. Given $\llbracket G, Y \rrbracket$, $IB(G_{\overline{W}}, Y)$ is a POMIS, for any $W \subseteq V \setminus \{Y\}$.



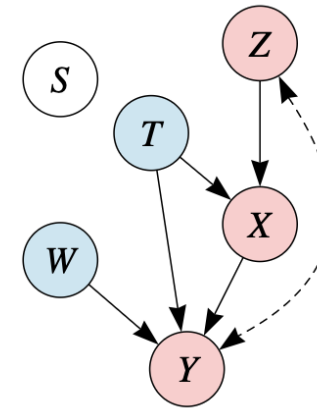
(a) G



(b) $G_{\overline{X}}$



(c) $G_{\overline{Z}}$



(d) $G_{\overline{W}}$

Graphical Characterization of POMIS

Intervention Border : Parents of MUCT

► Finding POMISs = Finding IBs

Proposition 4. $IB(G, Y)$ is a POMIS given $\llbracket G, Y \rrbracket$.

Proposition 5. Given $\llbracket G, Y \rrbracket$, $IB(G_{\overline{W}}, Y)$ is a POMIS, for any $W \subseteq V \setminus \{Y\}$.

Theorem 6. Given $\llbracket G, Y \rrbracket$, $X \subseteq V \setminus \{Y\}$ is a POMIS if and only if $IB(G_{\overline{X}}, Y) = X$.

► Naive Approach

- Enumerate every possible intervention set and get IB
- **DFS-like** approach will do the job ! ! !

Algorithmic Characterization of POMIS

Properties for finding POMISs

► DFS-like POMIS Search

Proposition 7. *Let $\mathbf{T}$ and $\mathbf{X}$ be the $\text{MUCT}(G_{\overline{\mathbf{W}}}, Y)$ and $\text{IB}(G_{\overline{\mathbf{W}}}, Y)$, respectively, relative to G and Y . Then, for any $\mathbf{Z} \subseteq \mathbf{V} \setminus \mathbf{T}$, $\text{MUCT}(G_{\overline{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{T}$ and $\text{IB}(G_{\overline{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{X}$.*

Proposition 8. *Let $H = G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ where $\mathbf{T}$ and $\mathbf{X}$ are MUCT and IB given $\llbracket G_{\overline{\mathbf{W}}}, Y \rrbracket$, respectively. Then, for any $\mathbf{W}' \subseteq \mathbf{T} \setminus \{\bar{Y}\}$, $H_{\overline{\mathbf{W}'}}$ and $G_{\overline{\mathbf{W} \cup \mathbf{W}'}}$ yield the same MUCT and IB with respect to Y .*

- We don't have to see outside of IB, and MUCT
 - Intervening outside variable does not affect Y (Rule 3 of Do-Calculus)
 - We don't have to see whole graph but a subgraph

Algorithmic Characterization of POMIS

Properties for finding POMISs

► DFS-like POMIS Search

Algorithm 1 Algorithm enumerating all POMISs with $\llbracket G, Y \rrbracket$

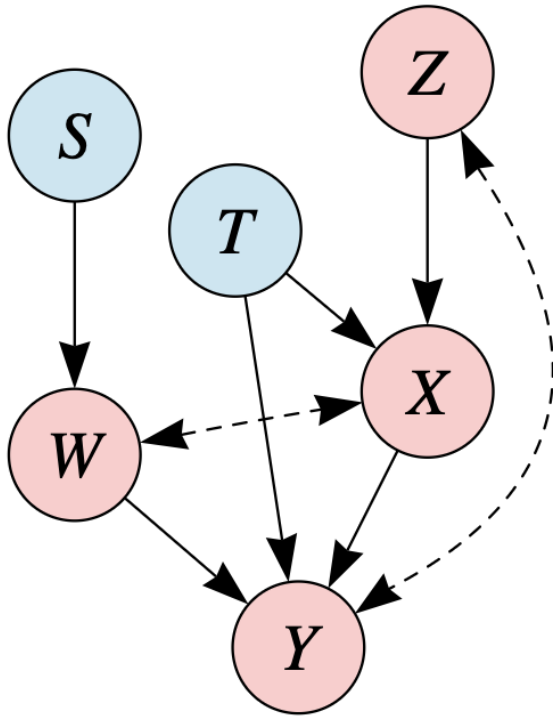
```
1: function POMISs( $G, Y$ )
2:    $\mathbf{T}, \mathbf{X} = \text{MUCT}(G, Y), \text{IB}(G, Y); H = G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ 
3:   return  $\{\mathbf{X}\} \cup \text{subPOMISs}(H, Y, \text{reversed}(\text{topological-sort}(H)) \cap (\mathbf{T} \setminus \{Y\}), \emptyset)$ 

4: function SUBPOMISs( $G, Y, \pi, \mathbf{O}$ )
5:    $\mathbf{P} = \emptyset$ 
6:   for  $\pi_i \in \pi$  do
7:      $\mathbf{T}, \mathbf{X}, \pi', \mathbf{O}' = \text{MUCT}(G_{\overline{\pi_i}}, Y), \text{IB}(G_{\overline{\pi_i}}, Y), \pi^{i+1:|\pi|} \cap \mathbf{T}, \mathbf{O} \cup \pi^{1:i-1}$ 
8:     if  $\mathbf{X} \cap \mathbf{O}' = \emptyset$  then
9:        $\mathbf{P} = \mathbf{P} \cup \{\mathbf{X}\} \cup (\text{subPOMISs}(G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}], Y, \pi', \mathbf{O}') \text{ if } \pi' \neq \emptyset \text{ else } \emptyset)$ 
10:  return  $\mathbf{P}$ 
```

Algorithmic Characterization of POMIS

Properties for finding POMISs

► DFS-like POMIS Search



Reversed topological order : $W < X < Z$

$\{W\} : \text{do}(S,T,W) - \text{IB} = \{W, T\}$

$\{W, X\} : \text{do}(S,T,W,X) - \text{IB} = \{W, T, X\}$

$\{W, X, Z\} : \text{do}(S,T,W,X,Z) - \text{IB} = \{W, T, X\}$

$\{X\} : \text{do}(S,T,X) - \text{IB} = \{W, T, X\}$

$\{X, Z\}$ is pruned (graph containing $\text{do}(W)$ is exhaustively used)

$\{Z\} : \text{do}(S,T,Z) - \text{IB} = \{W, T, X\}$

Experiments

Regret Comparison : UCB, KL-UCB, TS (with and without POMIS)

Experiments with Front-Door, Back-Door

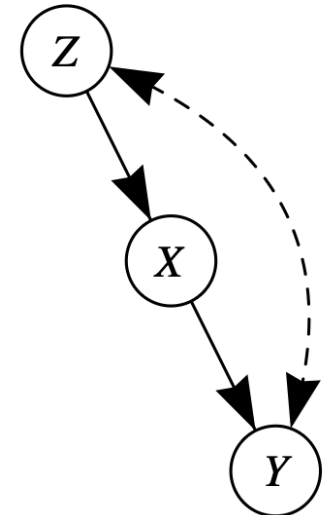
Environment

► Variables in the Graph are binary (0-1)

- Bernoulli Sampling for determining the value of node
- Weight is allocated for each edge just like Linear SCM
- Value of node is sampled from distribution where $P(V=1) = \frac{1}{1+e^{-w^T \cdot Pa(V)}}$
- 5 different random seeds were adopted

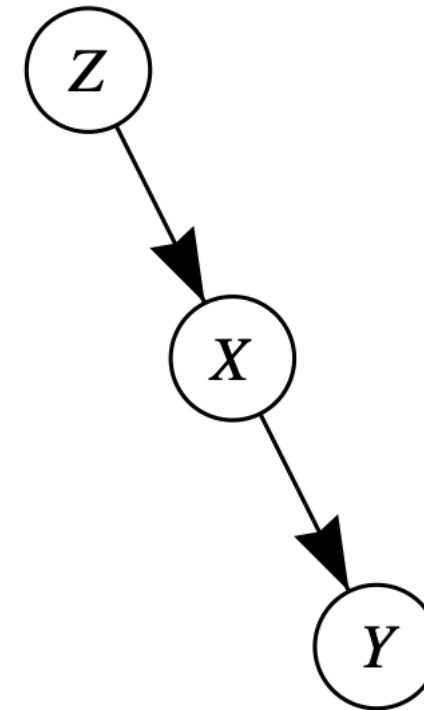
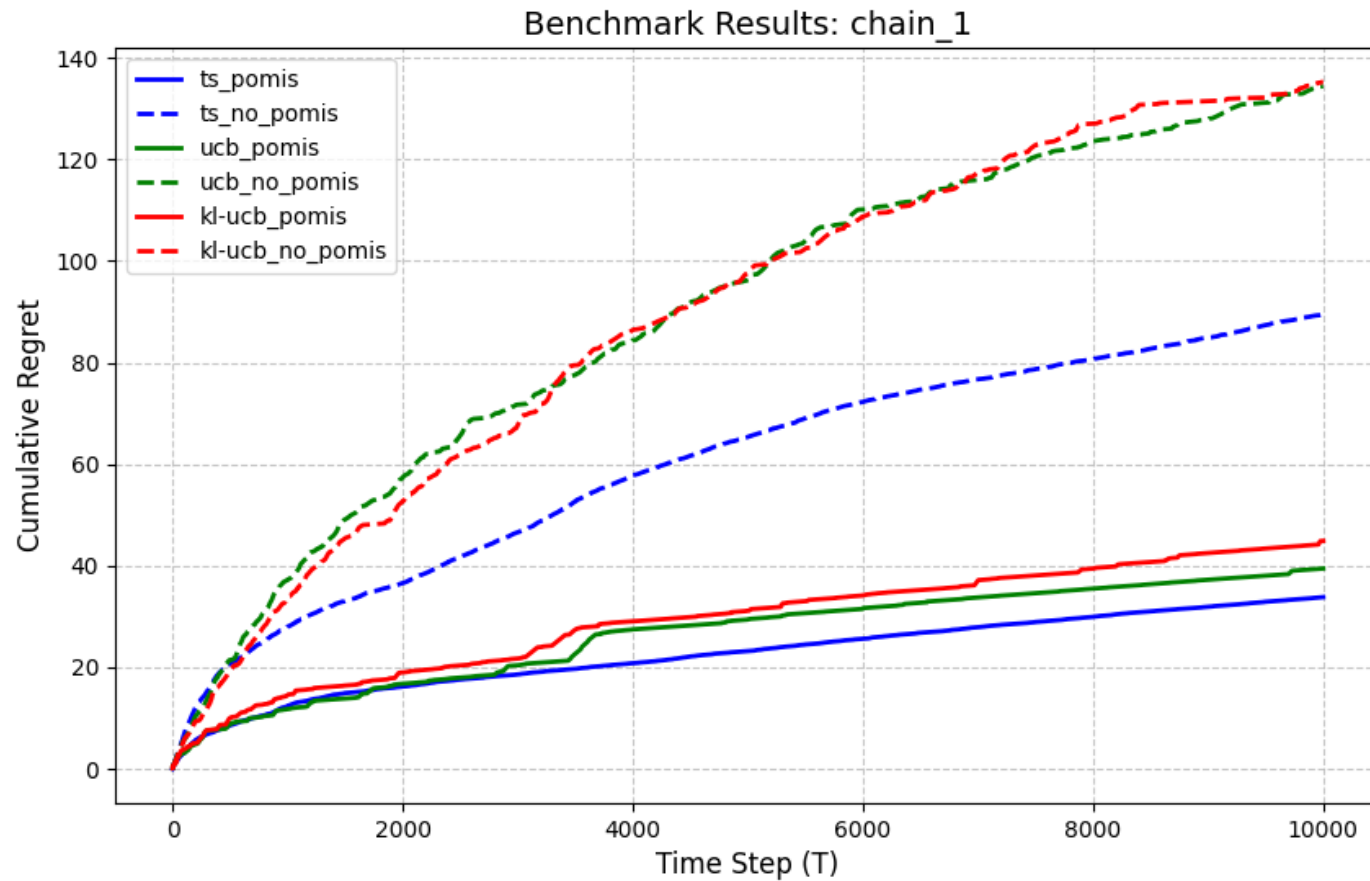
► We cannot know expected reward of each arm

- Use Monte Carlo Approximation (or sample-based estimation)
- We deem these estimated as oracle when we calculate the regret
- Arm we have to try is significantly reduced : $9 \rightarrow 3$



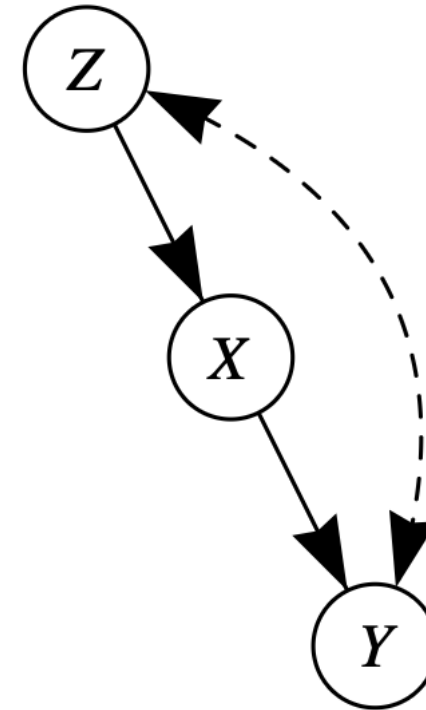
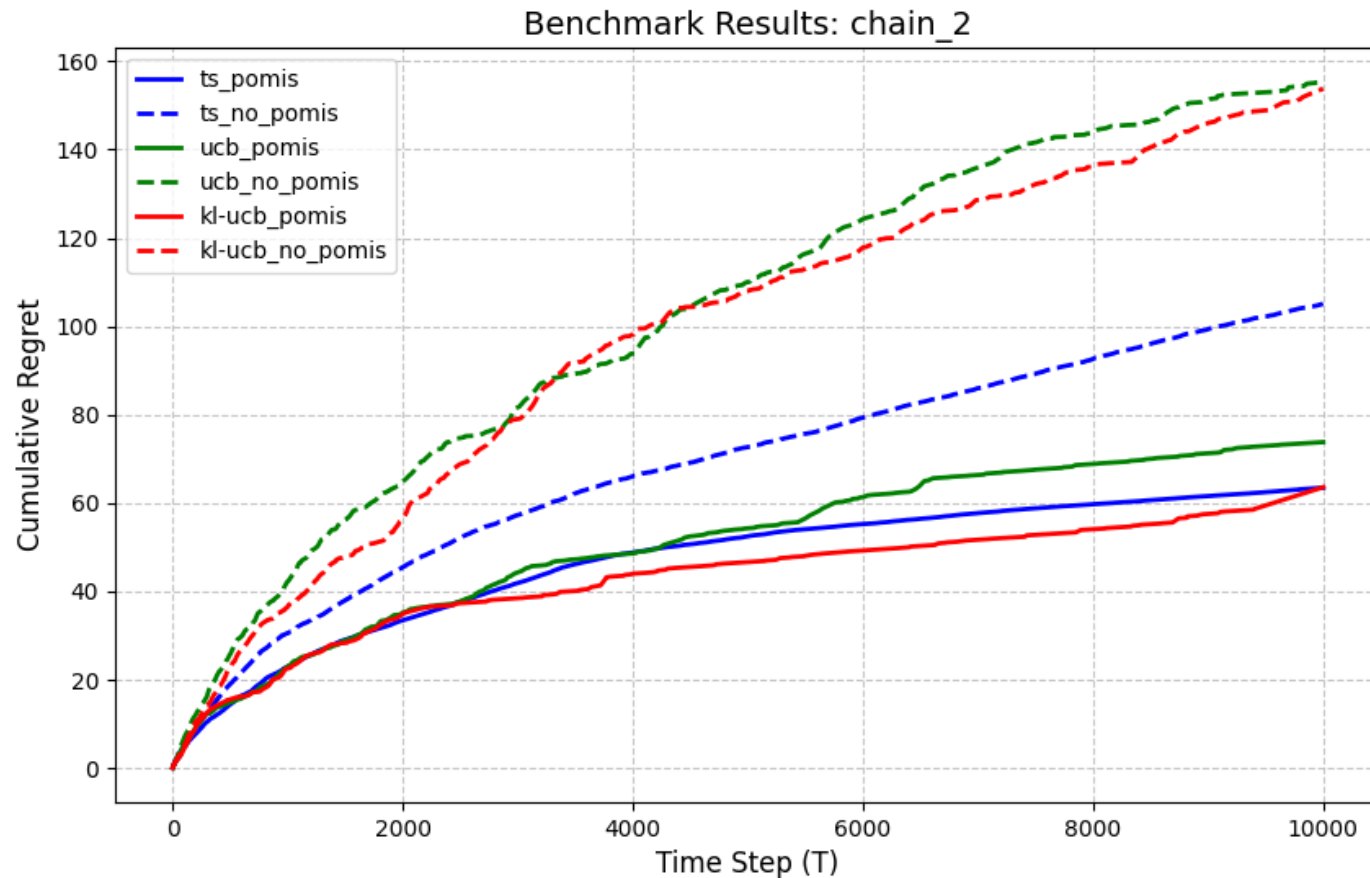
Experiments with UCB, KL-UCB, TS

Markovian Model



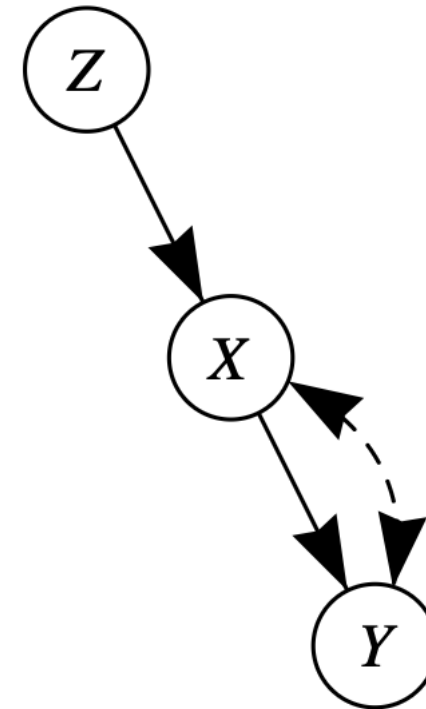
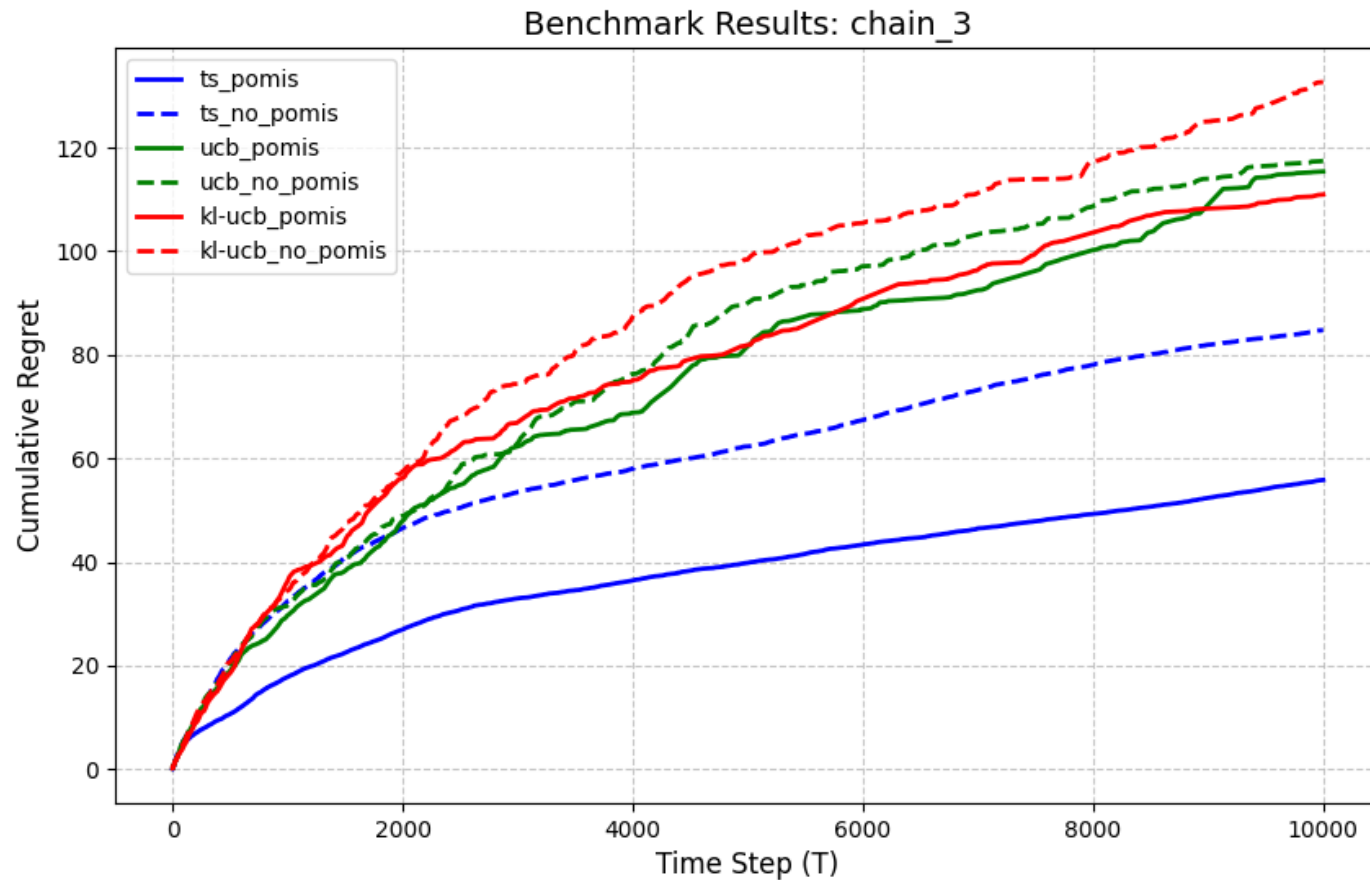
Experiments with UCB, KL-UCB, TS

Front-Door Case



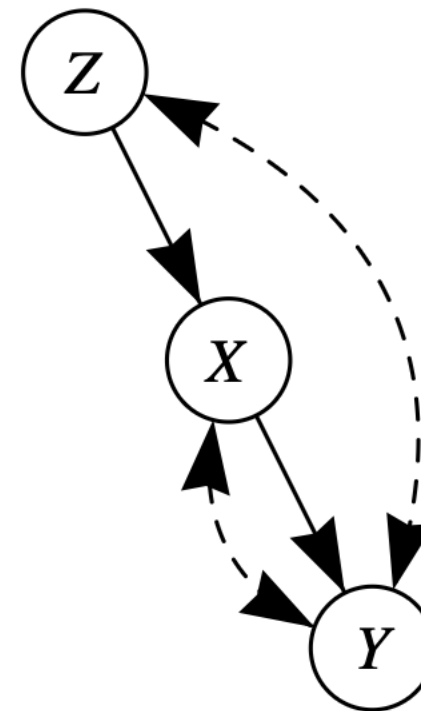
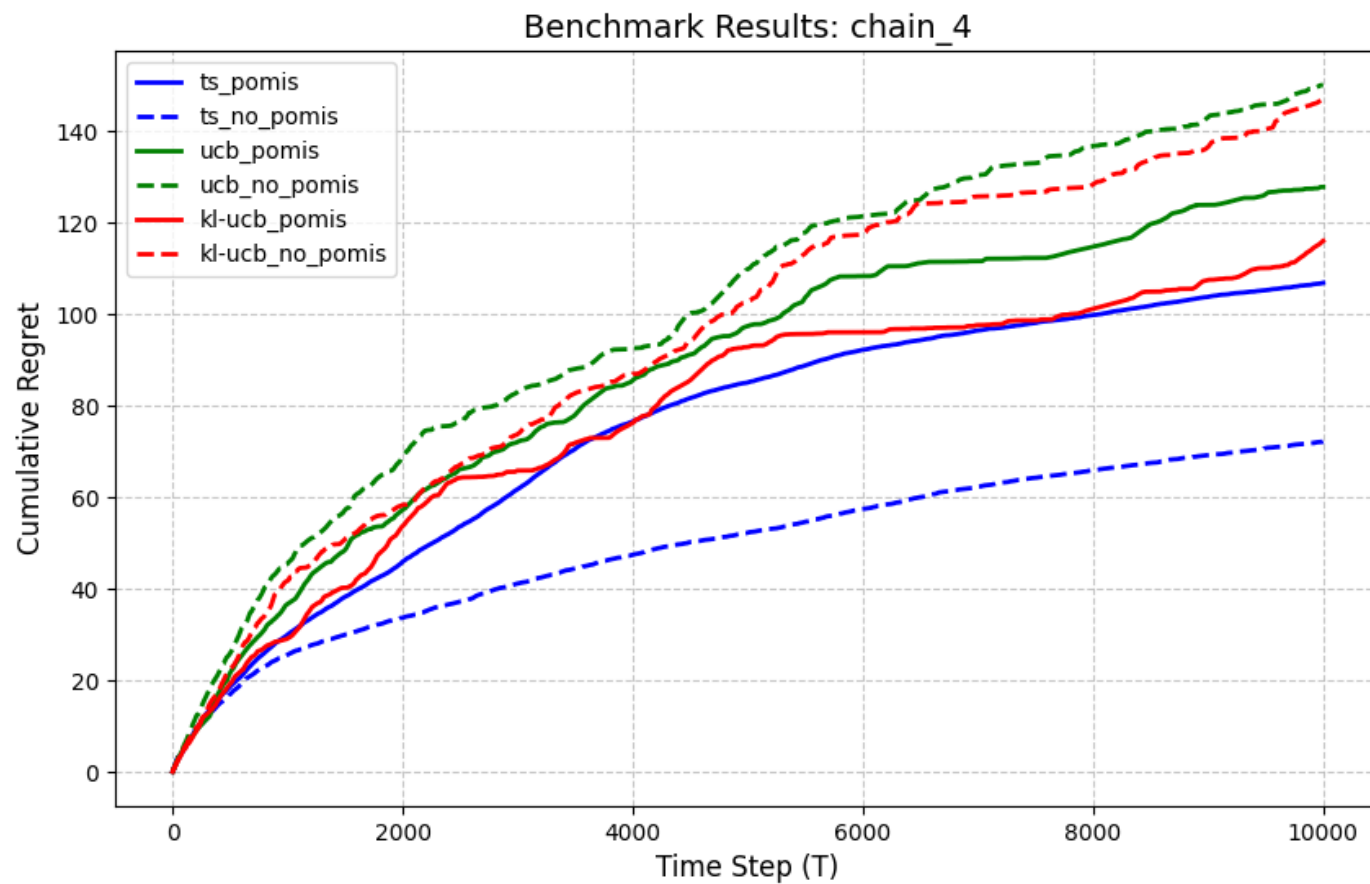
Experiments with UCB, KL-UCB, TS

IV Case



Experiments with UCB, KL-UCB, TS

Anomaly : Doubly Confounded Case



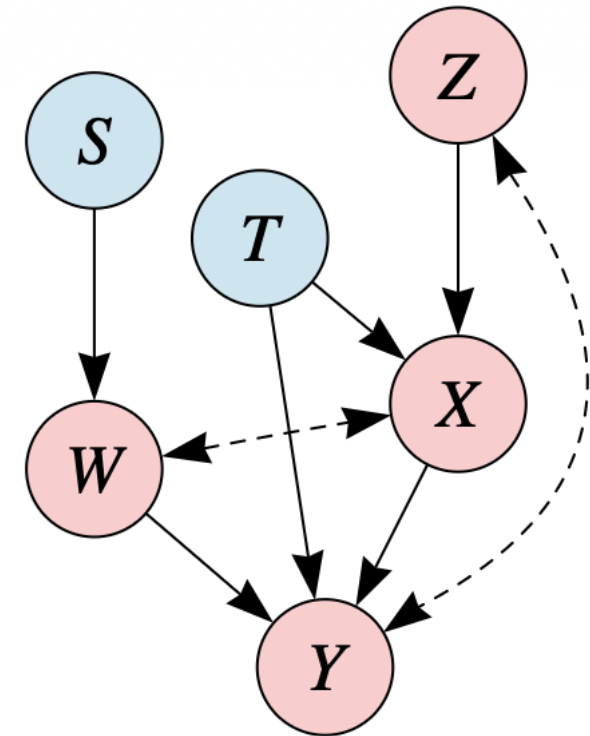
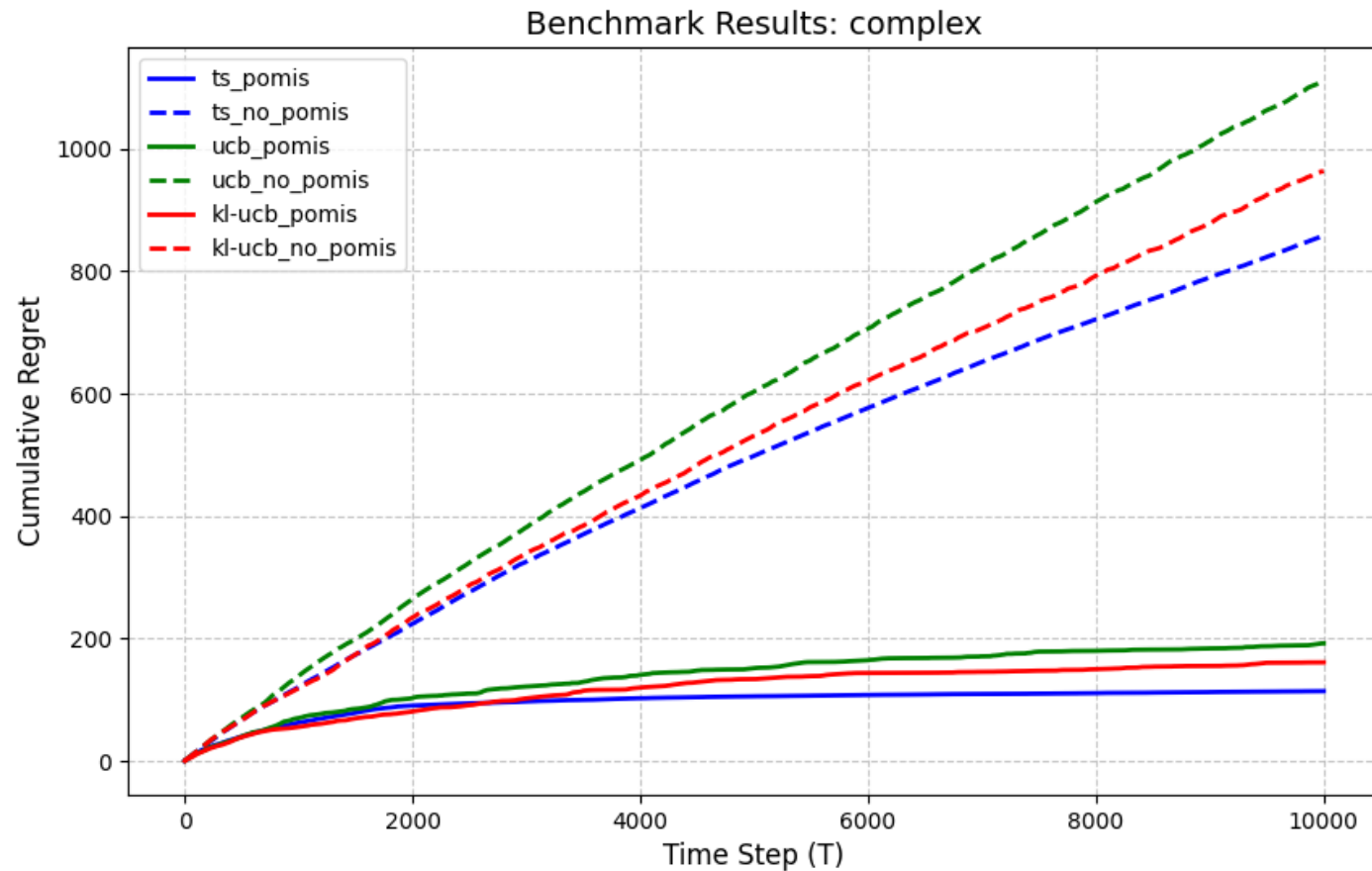
Anomaly Analysis

Thompson Sampling without POMIS Filtering

- ▶ Benefit from **Exploration Redundancy**
 - Multiple Optimal Arms are contained
 - In environment with high reward variance capturing one of optimal arms can be better strategy
- ▶ Risk Dispersion
 - POMISs reduces action space but only one optimal arm stands for it
 - Due to variance and bias, we might assume optimal arm suboptimal
 - However, without POMIS filtering, it works as an **Ensemble** so risk is dispersed

Experiments with UCB, KL-UCB, TS

Complex Case



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